

Appendix A. Additional Prior Work

Expression-based Prediction requires feature selection to improve interpretability and reduce overfitting, in addition to supervised prediction via classification or survival analysis. *Wrapper* and *filter* algorithms use a predictive model or model-agnostic measure to score features, respectively; for example, AtheroSpectrum used wrapper-style search over random gene subsets to construct the CR-30 signature and then integrated this signature with clinical covariates¹. Wrapper methods are often better aligned with predictive objectives but are computationally expensive and can be unstable in high-dimensional regimes, whereas filter methods are computationally efficient but can be weakly coupled to the final prediction task. Embedded methods, such as L1-penalized logistic regression, optimize prediction and feature sparsity jointly and are therefore widely used in transcriptomic prediction pipelines. Because of their interpretability, generalized linear models are widely adopted to estimate disease risk^{2–4}; for instance, the coefficients of a logistic regression provide an interpretation of how gene expression changes affect the disease risk log-odds. Machine learning models like random forests or boosting classifiers have also been used for disease prediction^{5,6} and allow for learning interactions between features⁷, though these approaches are generally considered less interpretable than linear models and thus potentially less appropriate for clinical use⁸. Further, since machine learning models typically require more samples than linear regressors and most prior expression-based prediction studies have small sample sizes, gene expression is often imputed from genetic or epigenetic data⁹; however, gene expression is difficult to predict from genomics data¹⁰, with a small proportion of genes being able to be predicted with $r^2 > 0.1$ ^{11,12}. Across these approaches, the simplifying assumptions of additive linear effects^{4,13} or unstructured interactions among gene expression features^{5,6} poorly align with the coordinated gene activity driving disease-relevant cellular processes.

Appendix B. Variational Inference

The Poisson factorization, spike-and-slab structure, and supervised logistic term produce a coupled posterior over all latent variables, so exact inference is intractable. We use mean-field variational inference with coordinate-ascent updates, rendering the supervised term conditionally quadratic by Pólya–Gamma (PG) data augmentation¹⁴.

Appendix B.1. Pólya–Gamma Augmentation of the Supervised Term.

The only non-conjugate component of the model is the Bernoulli supervision

$$p(y_{jk} \mid \theta_j, v_k, \gamma_k, \mathbf{x}_j^{\text{aux}}) = \sigma(A_{jk})^{y_{jk}} (1 - \sigma(A_{jk}))^{1-y_{jk}}, \quad A_{jk} = \theta_j v_k^\top + \mathbf{x}_j^{\text{aux}} \gamma_k^\top,$$

where σ is the logistic function and $\theta_j, v_k, \mathbf{x}_j^{\text{aux}}, \gamma_k$ are row vectors. Following Polson, Scott, and Windle¹⁴, for a Pólya–Gamma variable $\zeta \sim \text{PG}(1, 0)$,

$$\frac{(e^A)^y}{1 + e^A} = \frac{1}{2} e^{\tilde{\kappa} A} \int_0^\infty e^{-\zeta A^2/2} p_{\text{PG}}(\zeta \mid 1, 0) d\zeta, \quad \tilde{\kappa} = y - \frac{1}{2}. \quad (\text{B.1})$$

Introducing one auxiliary variable ζ_{jk} per sample–phenotype pair gives the augmented likeli-

hood

$$p(y_{jk}, \zeta_{jk} \mid A_{jk}) \propto \exp\left(\tilde{\kappa}_{jk} A_{jk} - \frac{\zeta_{jk}}{2} A_{jk}^2\right) p_{\text{PG}}(\zeta_{jk} \mid 1, 0), \quad \tilde{\kappa}_{jk} = y_{jk} - \frac{1}{2}. \quad (\text{B.2})$$

We write the PG shift as $\tilde{\kappa}_{jk}$ to distinguish it from the number of phenotypes κ . Equation (B.1) is an exact identity rather than a variational bound, so the augmented model has the same marginal likelihood as the original. Conditional on ζ_{jk} the exponent is quadratic in A_{jk} , which is conjugate to the Gaussian factors $q(v_k)$ and $q(\gamma_k)$ but only log-quadratic in θ_j ; the consequences for the θ update are given in the remark below.

Appendix B.2. Joint Distribution and Variational Family.

Conditioned on the auxiliary covariates and including the augmentation variables, the joint distribution factorizes as

$$\begin{aligned} p(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}, \boldsymbol{\xi}, \tilde{\boldsymbol{\beta}}, \boldsymbol{\pi}, \boldsymbol{\eta}, \mathbf{v}, \mathbf{s}, \boldsymbol{\gamma}, \boldsymbol{\zeta} \mid \mathbf{X}^{\text{aux}}, M) &= \prod_{g=1}^p \left[p(\pi_g) p(\eta_g) \prod_{\ell=1}^d p(m_{g\ell} \mid \pi_g) p(\tilde{\beta}_{g\ell} \mid \eta_g) \right] \\ &\times \prod_{j=1}^n \left[p(\xi_j) \prod_{\ell=1}^d p(\theta_{j\ell} \mid \xi_j) \prod_{g=1}^p p(x_{jg} \mid \theta_j, \beta_g) \right] \\ &\times \prod_{k=1}^{\kappa} \left[p(\gamma_k) \prod_{\ell=1}^d p(s_{k\ell}) p(v_{k\ell} \mid s_{k\ell}) \prod_{j=1}^n p(y_{jk}, \zeta_{jk} \mid \theta_j, v_k, \gamma_k, \mathbf{x}_j^{\text{aux}}) \right]. \end{aligned}$$

We assume the fully factorized variational family

$$\begin{aligned} q(\eta_g) &= \text{Gamma}(a_{\eta_g}, b_{\eta_g}), & q(\tilde{\beta}_{g\ell}) &= \text{Gamma}(a_{\tilde{\beta}_{g\ell}}, b_{\tilde{\beta}_{g\ell}}), \\ q(m_{g\ell}) &= \text{Bernoulli}(\rho_{g\ell}), & q(\pi_g) &= \text{Beta}(a_{\pi_g}, b_{\pi_g}), \\ q(\xi_j) &= \text{Gamma}(a_{\xi_j}, b_{\xi_j}), & q(\theta_{j\ell}) &= \text{Gamma}(a_{\theta_{j\ell}}, b_{\theta_{j\ell}}), \\ q(v_{k\ell}) &= \mathcal{N}(\mu_{v_{k\ell}}, \tau_{v_{k\ell}}^2), & q(s_{k\ell}^{-1}) &= \text{InvGaussian}(\mu_{s_{k\ell}}, \lambda_{s_{k\ell}}), \\ q(\gamma_k) &= \mathcal{N}(\mu_{\gamma_k}, \Sigma_{\gamma_k}), & q(\zeta_{jk}) &= \text{PG}(1, c_{jk}), \end{aligned}$$

where the Bayesian-Lasso factor is placed on the *precision* $s_{k\ell}^{-1}$, which is inverse-Gaussian under the Gaussian-exponential scale mixture. For any Gamma factor $q(X) = \text{Gamma}(a, b)$ we use $\mathbb{E}_q[X] = a/b$, $\mathbb{E}_q[\log X] = \Psi(a) - \log b$, and $\text{Var}_q(X) = a/b^2$, where $\Psi(\cdot)$ is the digamma function.

Optimal augmentation factor. Because the augmented exponent in (B.2) is $-\frac{1}{2}\zeta_{jk}A_{jk}^2$, the coordinate-optimal factor for ζ_{jk} is again Pólya-Gamma, exponentially tilted with $c_{jk} = \sqrt{\mathbb{E}_q[A_{jk}^2]}$. The only moment used downstream is its mean,

$$\bar{\zeta}_{jk} \equiv \mathbb{E}_q[\zeta_{jk}] = \frac{1}{2c_{jk}} \tanh\left(\frac{c_{jk}}{2}\right), \quad \lim_{c_{jk} \rightarrow 0} \bar{\zeta}_{jk} = \frac{1}{4}, \quad (\text{B.3})$$

evaluated by a second-order Taylor expansion of $\tanh$ when c_{jk} is small.

Count allocation and gene-side updates. Following Raikov's decomposition theorem¹⁵, we introduce latent allocations $z_{jg\ell}$ with $x_{jg} = \sum_{\ell=1}^d z_{jg\ell}$ and $z_{jg\ell} \sim \text{Poisson}(\theta_{j\ell}\beta_{g\ell})$. Because

$\beta_{g\ell} = m_{g\ell}\tilde{\beta}_{g\ell}$, inactive gene-program pairs contribute zero rate automatically. Writing $e_{jg\ell} = \exp(\Psi(a_{\theta_{j\ell}}) - \log b_{\theta_{j\ell}} + \Psi(a_{\tilde{\beta}_{g\ell}}) - \log b_{\tilde{\beta}_{g\ell}})$, the optimal multinomial allocation is

$$\phi_{jg\ell} = \frac{\rho_{g\ell} e_{jg\ell}}{\sum_{\ell'=1}^d \rho_{g\ell'} e_{jg\ell'}}, \quad \mathbb{E}_q[z_{jg\ell}] = x_{jg}\phi_{jg\ell}, \quad (\text{B.4})$$

and the slab loading update is

$$a_{\tilde{\beta}_{g\ell}} = \alpha_\beta + \sum_{j=1}^n x_{jg}\phi_{jg\ell}, \quad b_{\tilde{\beta}_{g\ell}} = \mathbb{E}_q[\eta_g] + \rho_{g\ell} \sum_{j=1}^n \mathbb{E}_q[\theta_{j\ell}]. \quad (\text{B.5})$$

For the Bernoulli membership variable, the log-odds update compares the expected log-joint under $m_{g\ell} = 1$ and $m_{g\ell} = 0$. Let $\bar{\phi}_{jg\ell} = e_{jg\ell}(e_{jg\ell} + \sum_{m \neq \ell} \rho_{gm} e_{jgm})^{-1}$ be the allocation with program ℓ held active, and define

$$L_{jg\ell} = x_{jg} \bar{\phi}_{jg\ell} \left(\Psi(a_{\theta_{j\ell}}) - \log b_{\theta_{j\ell}} + \Psi(a_{\tilde{\beta}_{g\ell}}) - \log b_{\tilde{\beta}_{g\ell}} \right) - \mathbb{E}_q[\theta_{j\ell}] \mathbb{E}_q[\tilde{\beta}_{g\ell}] - \log \Gamma(x_{jg} \bar{\phi}_{jg\ell} + 1);$$

the final term evaluates $\mathbb{E}_q[\log z_{jg\ell}!]$ at $\mathbb{E}_q[z_{jg\ell}]$ rather than exactly. The update is

$$\log \frac{\rho_{g\ell}}{1 - \rho_{g\ell}} = \Psi(a_{\pi_g}) - \Psi(b_{\pi_g}) + \sum_{j=1}^n L_{jg\ell}. \quad (\text{B.6})$$

The conjugate hyperparameter updates are

$$a_{\pi_g} = \alpha_\pi + \sum_{\ell=1}^d \rho_{g\ell}, \quad b_{\pi_g} = \beta_\pi + d - \sum_{\ell=1}^d \rho_{g\ell}, \quad (\text{B.7})$$

$$a_{\eta_g} = \alpha_\eta + d\alpha_\beta, \quad b_{\eta_g} = \lambda_\eta + \sum_{\ell=1}^d \frac{a_{\tilde{\beta}_{g\ell}}}{b_{\tilde{\beta}_{g\ell}}}, \quad a_{\xi_j} = \alpha_\xi + d\alpha_\theta, \quad b_{\xi_j} = \lambda_\xi + \sum_{\ell=1}^d \frac{a_{\theta_{j\ell}}}{b_{\theta_{j\ell}}}. \quad (\text{B.8})$$

Pathway mask. In the *de novo* setting all gene-program pairs are updated freely. In the pathway-guided setting we fix $m_{g\ell} = 0$, $\rho_{g\ell} = 0$, and $\beta_{g\ell} = 0$ for every entry with $M_{g\ell} = 0$, which implies $\phi_{jg\ell} = 0$ for all j ; the same updates are applied to the allowed entries only. The gene-side sums over ℓ in (B.7)–(B.8) run over allowed entries, so d is replaced by $d_g = \sum_\ell M_{g\ell}$ in those three updates. The sample-side count d in a_{ξ_j} is unchanged, since θ is not masked.

Supervised updates under augmentation. Conditional on $q(\zeta)$, the augmented supervised term (B.2) is quadratic in A_{jk} , and $\bar{\zeta}_{jk}$ plays the role of the quadratic-curvature coefficient. Throughout we use the leave-one-out predictor $C_{jk}^{(-\ell)} = \sum_{m \neq \ell} \mathbb{E}_q[\theta_{jm}] \mathbb{E}_q[v_{km}] + \mathbf{x}_j^{\text{aux}} \mu_{\gamma_k}$, i.e. A_{jk} with the ℓ -th program term removed. The sample-activity update is

$$a_{\theta_{j\ell}} = \alpha_\theta + \sum_{g=1}^p x_{jg}\phi_{jg\ell}, \quad b_{\theta_{j\ell}} = \mathbb{E}_q[\xi_j] + \sum_{g=1}^p \rho_{g\ell} \mathbb{E}_q[\tilde{\beta}_{g\ell}] + R_{j\ell}, \quad (\text{B.9})$$

$$R_{j\ell} = \sum_{k=1}^{\kappa} \left[-\bar{\kappa}_{jk} \mathbb{E}_q[v_{k\ell}] + \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}] C_{jk}^{(-\ell)} + \frac{1}{2} \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}^2] \mathbb{E}_q[\theta_{j\ell}] \right]. \quad (\text{B.10})$$

The supervised correction $R_{j\ell}$ enters the Gamma rate of $\theta_{j\ell}$ and couples program activity to disease status: the label-linear term shifts disease-relevant programs ($v_{k\ell} \neq 0$) toward higher

activity in cases and lower activity in controls, while the $\bar{\zeta}_{jk}$ -scaled terms damp that shift by the curvature of the logistic.

Remark (exactness of the θ update). Conditional on ζ , the terms in $\theta_{j\ell}$ contributed by (B.2) are $\tilde{\kappa}_{jk} v_{k\ell} \theta_{j\ell} - \bar{\zeta}_{jk} v_{k\ell} C_{jk}^{(-\ell)} \theta_{j\ell} - \frac{1}{2} \bar{\zeta}_{jk} v_{k\ell}^2 \theta_{j\ell}^2$, so the coordinate-optimal factor is $q^*(\theta_{j\ell}) \propto \theta_{j\ell}^{a-1} \exp(-b\theta_{j\ell} - c\theta_{j\ell}^2)$ with $c = \frac{1}{2} \sum_k \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}^2]$. The quadratic term lies outside the span of the Gamma sufficient statistics $(\log \theta, \theta)$, so q^* is not Gamma. Equation (B.9) therefore restricts $q(\theta_{j\ell})$ to the Gamma family and linearizes $c\theta_{j\ell}^2$ about the current mean, replacing it with $(c\mathbb{E}_q[\theta_{j\ell}])\theta_{j\ell}$; the resulting update is a fixed-point iteration rather than an exact coordinate maximizer. Because the linearized rate may be negative, it is floored at a small $\varepsilon > 0$, $b_{\theta_{j\ell}} \leftarrow \max(b_{\theta_{j\ell}}, \varepsilon)$, which is a projection back into the family.

Regression and scale updates. The regression coefficient updates are

$$\tau_{v_{k\ell}}^{-2} = \mathbb{E}_q[s_{k\ell}^{-1}] + \sum_{j=1}^n \bar{\zeta}_{jk} \mathbb{E}_q[\theta_{j\ell}^2], \quad \mu_{v_{k\ell}} = \tau_{v_{k\ell}}^2 \sum_{j=1}^n [\tilde{\kappa}_{jk} - \bar{\zeta}_{jk} C_{jk}^{(-\ell)}] \mathbb{E}_q[\theta_{j\ell}], \quad (\text{B.11})$$

with $\mathbb{E}_q[\theta_{j\ell}^2] = \mathbb{E}_q[\theta_{j\ell}]^2 + \text{Var}_q(\theta_{j\ell})$, and

$$\Sigma_{\gamma_k}^{-1} = \frac{1}{\sigma_{\gamma}^2} I + \sum_{j=1}^n \bar{\zeta}_{jk} (\mathbf{x}_j^{\text{aux}})^{\top} \mathbf{x}_j^{\text{aux}}, \quad \mu_{\gamma_k} = \Sigma_{\gamma_k} \sum_{j=1}^n (\mathbf{x}_j^{\text{aux}})^{\top} [\tilde{\kappa}_{jk} - \bar{\zeta}_{jk} \mathbb{E}_q[\theta_j] \mathbb{E}_q[v_k]^{\top}]. \quad (\text{B.12})$$

The Bayesian-Lasso scale update is unchanged by the augmentation, since the prior on v is untouched. Under $v_{k\ell} | s_{k\ell} \sim \mathcal{N}(0, s_{k\ell})$, $s_{k\ell} \sim \text{Exp}(1/2b_v^2)$, the optimal factor for $s_{k\ell}^{-1}$ is inverse-Gaussian with $\lambda_{s_{k\ell}} = b_v^{-2}$ and

$$\mathbb{E}_q[s_{k\ell}^{-1}] = \mu_{s_{k\ell}} = \frac{1}{b_v \sqrt{\mathbb{E}_q[v_{k\ell}^2]}}. \quad (\text{B.13})$$

Finally, the tilt is refreshed by setting $c_{jk}^2 = \mathbb{E}_q[A_{jk}^2]$. Because the variational factors are independent, $\text{Var}_q(A_{jk})$ decomposes into separate θ -, v -, and γ -contributions,

$$\begin{aligned} \mathbb{E}_q[A_{jk}] &= \mathbb{E}_q[\theta_j] \mu_{v_k}^{\top} + \mathbf{x}_j^{\text{aux}} \mu_{\gamma_k}, \\ c_{jk}^2 &= \mathbb{E}_q[A_{jk}^2] = \mathbb{E}_q[A_{jk}]^2 + \sum_{\ell=1}^d [\text{Var}_q(\theta_{j\ell}) \mathbb{E}_q[v_{k\ell}^2] + \mathbb{E}_q[\theta_{j\ell}]^2 \tau_{v_{k\ell}}^2] + \mathbf{x}_j^{\text{aux}} \Sigma_{\gamma_k} (\mathbf{x}_j^{\text{aux}})^{\top}, \end{aligned} \quad (\text{B.14})$$

with $\mathbb{E}_q[v_{k\ell}^2] = \mu_{v_{k\ell}}^2 + \tau_{v_{k\ell}}^2$, after which $\bar{\zeta}_{jk}$ is recomputed via (B.3). All three variance terms are required for the tilt to equal the full second moment, so that the residual in (B.15) vanishes at the optimum.

Objective and optimization. The variational objective is the evidence lower bound of the augmented model, $\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}, \boldsymbol{\xi}, \tilde{\boldsymbol{\beta}}, \mathbf{M}, \boldsymbol{\pi}, \boldsymbol{\eta}, \mathbf{v}, \mathbf{s}, \boldsymbol{\gamma}, \boldsymbol{\zeta} | \mathbf{X}^{\text{aux}})] - \mathbb{E}_q[\log q]$, combining the expected log-priors, the augmented Poisson likelihood, the augmented supervised term, and the entropies of all factors. Marginalizing the optimal PG factor analytically, the supervised contribution at $q(\zeta_{jk}) = \text{PG}(1, c_{jk})$ reduces to

$$\mathcal{L}^{\text{sup}} = \sum_{j=1}^n \sum_{k=1}^{\kappa} \left[\tilde{\kappa}_{jk} \mathbb{E}_q[A_{jk}] - \frac{c_{jk}}{2} + \log \sigma(c_{jk}) - \frac{\bar{\zeta}_{jk}}{2} (\mathbb{E}_q[A_{jk}^2] - c_{jk}^2) \right], \quad (\text{B.15})$$

whose last term vanishes when the tilt is at its optimum $c_{jk}^2 = \mathbb{E}_q[A_{jk}^2]$. Because (B.1) is exact rather than a bound, $\mathcal{L}(q)$ is a valid lower bound on $\log p(\mathbf{X}, \mathbf{Y} \mid \mathbf{X}^{\text{aux}})$ at every iteration. Monotonicity holds for every block whose update is an exact coordinate maximizer, but not for θ , whose update is linearized; we therefore monitor the ELBO empirically and base model selection and early stopping on validation performance, with all reported predictive results evaluated on held-out data. Algorithm 2 summarizes the procedure.

Complexity. Let $\text{nnz}(\mathbf{X}) = |\{(j, g) : x_{jg} \neq 0\}|$. The allocation (B.4) is evaluated on the fly for each nonzero and never materialized, giving $O(\text{nnz}(\mathbf{X})d)$ time and $O(d)$ working memory per entry; the loading update (B.5) reuses the same sums. In (B.6) the term $\mathbb{E}_q[\theta_{j\ell}]\mathbb{E}_q[\tilde{\beta}_{g\ell}]$ is nonzero when $x_{jg} = 0$, but factorizes as $(\sum_j \mathbb{E}_q[\theta_{j\ell}])\mathbb{E}_q[\tilde{\beta}_{g\ell}]$, so its cost is $O(pd)$ rather than $O(npd)$. Accumulating $\sum_m \mathbb{E}_q[\theta_{jm}]\mathbb{E}_q[v_{km}]$ once per (j, k) and subtracting the ℓ -th term makes all $C_{jk}^{(-\ell)}$ available in $O(nd\kappa)$ rather than $O(nd^2\kappa)$, which also bounds (B.11) and (B.14); the auxiliary update (B.12) costs $O(n(p^{\text{aux}})^2 + (p^{\text{aux}})^3)$ per phenotype. A training sweep is therefore $O(\text{nnz}(\mathbf{X})d + nd\kappa + \kappa\{n(p^{\text{aux}})^2 + (p^{\text{aux}})^3\})$, dominated by the first term whenever $p^{\text{aux}} \ll p$. Memory is $O(\text{nnz}(\mathbf{X}) + (n + p)d + \kappa(d + (p^{\text{aux}})^2))$. Fold-in for one held-out sample costs $O(\text{nnz}(\mathbf{x}_{j^*})d)$ per local iteration and touches no global parameter, so test-time inference is embarrassingly parallel across samples.

Test- and validation-time inference (fold-in). At test time the labels y_{j^*k} are unobserved, so (B.2) contributes no likelihood factor for a new sample j^* . We freeze the *global* parameters—the gene-side factors $\{a_{\tilde{\beta}}, b_{\tilde{\beta}}, \rho, a_{\eta}, b_{\eta}, a_{\pi}, b_{\pi}\}$ and the regression coefficients $\{\mu_v, \tau_v^2, \mu_{\gamma}, \Sigma_{\gamma}\}$ —and infer only the *local* variables $\{\theta_{j^*\ell}, \xi_{j^*}\}$ from the counts x_{j^*g} through the allocations $\phi_{j^*g\ell}$. The supervised correction vanishes, $R_{j^*\ell} = 0$, no PG variables ζ_{j^*k} are needed, and (B.9) reduces to its unsupervised form

$$a_{\theta_{j^*\ell}} = \alpha_{\theta} + \sum_{g=1}^p x_{j^*g} \phi_{j^*g\ell}, \quad b_{\theta_{j^*\ell}} = \mathbb{E}_q[\xi_{j^*}] + \sum_{g=1}^p \rho_{g\ell} \mathbb{E}_q[\tilde{\beta}_{g\ell}]. \quad (\text{B.16})$$

Note that θ_{j^*} is thus driven only by held-out expression and the fixed gene programs, whereas training activities are label-informed through $R_{j\ell}$. Given the converged fold-in posterior, the predictor moments are read off from (B.14) as $\mathbb{E}_q[A_{j^*k}] = \mathbb{E}_q[\theta_{j^*}]\mu_{v_k}^{\top} + \mathbf{x}_{j^*}^{\text{aux}}\mu_{\gamma_k}$ and $\text{Var}_q(A_{j^*k}) = c_{j^*k}^2 - \mathbb{E}_q[A_{j^*k}]^2$, and the posterior predictive probability is approximated by the Gaussian–logistic integral

$$\hat{y}_{j^*k} = \mathbb{E}_q[\sigma(A_{j^*k})] \approx \sigma\left(\frac{\mathbb{E}_q[A_{j^*k}]}{\sqrt{1 + \pi \text{Var}_q(A_{j^*k})/8}}\right), \quad (\text{B.17})$$

where π denotes the mathematical constant. This reduces to the plug-in estimate $\sigma(\mathbb{E}_q[A_{j^*k}])$ in the zero-variance limit. The same routine supplies the validation predictions used for model selection in Algorithm 2, and is summarized in Algorithm 3.

Algorithm 2 Pólya–Gamma CAVI for DRGP

Input: Expression $\mathbf{X}$, labels $\mathbf{Y}$, auxiliary covariates $\mathbf{X}^{\text{aux}}$, number of factors d , pathway mask $\mathbf{M}$, and model hyperparameters

Output: Fitted global and local variational parameters

- 1: Initialize all variational parameters and enforce $\mathbf{M}$
 - 2: **while** the ELBO has not converged **do**
 - 3: Update count allocations ϕ ▷ Eq. (B.4)
 - 4: Update gene loadings and memberships $(\tilde{\beta}, \rho)$ ▷ Eqs. (B.5), (B.6)
 - 5: Update gene-level parameters (π, η) ▷ Eq. (B.7), (B.8)
 - 6: Update sample activities (θ, ξ) ▷ Eqs. (B.9), (B.8)
 - 7: Update supervised parameters (s, v, γ) ▷ Eqs. (B.11)–(B.13)
 - 8: Update Pólya–Gamma moments c and $\bar{\zeta}$ ▷ Eqs. (B.14), (B.3)
 - 9: **end while**
 - 10: **return** fitted variational parameters
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Algorithm 3 Fold-in inference and risk prediction

Input: Held-out expression $\mathbf{x}_{j^*}$, auxiliary covariates $\mathbf{x}_{j^*}^{\text{aux}}$, and frozen global parameters from Algorithm 2

Output: Predicted risks $\hat{y}_{j^*k}$

- 1: **for** each held-out sample j^* **do**
 - 2: Initialize local parameters $(\phi_{j^*}, \theta_{j^*}, \xi_{j^*})$
 - 3: **while** the local Poisson objective has not converged **do**
 - 4: Update ϕ_{j^*} , then θ_{j^*} with $R_{j^*\ell} = 0$, then ξ_{j^*} ▷ Eqs. (B.4), (B.16), (B.8)
 - 5: **end while**
 - 6: **for** each phenotype k **do**
 - 7: Compute posterior moments of A_{j^*k} and predict $\hat{y}_{j^*k}$ ▷ Eqs. (B.14), (B.17)
 - 8: **end for**
 - 9: **end for**
 - 10: **return** $\{\hat{y}_{j^*k}\}$
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Appendix C. Ranking Genes within DRGP Programs

For all downstream analyses requiring a continuous ranking of genes within a DRGP program, we use

$$\hat{s}_{g\ell} = \mathbb{E}_q[\beta_{g\ell}] \mathbb{E}_q[\eta_g]. \quad (\text{C.1})$$

The expected loading $\mathbb{E}_q[\beta_{g\ell}]$ incorporates the posterior gene–program inclusion probability, while $\mathbb{E}_q[\eta_g]$ adjusts for gene-specific loading capacity. Consequently, $\hat{s}_{g\ell}$ favors genes that load strongly on program ℓ but do not load broadly across many programs.

For biological program summaries, genes were ordered by $\hat{s}_{g\ell}$ and the top 50 genes were retained where a fixed-size list was required. In simulation, support recovery was evaluated using $\hat{s}_{\cdot\ell}$ as the continuous gene score, with fitted programs Hungarian-matched to simulated programs by signed cosine similarity of the gene-score vectors. Program reproducibility across

random initializations was likewise evaluated by Jaccard similarity between gtop-50 gene sets.

Appendix D. Hyperparameter selection

For the program-recovery and prediction simulations, hyperparameters were fixed before evaluation and held constant across all simulated difficulty settings. This design prevents changes in model tuning from being confounded with changes in perturbation strength, responder-cell prevalence, or the transcriptomic and auxiliary liability shares. Both simulation arms used $\alpha_\theta = \alpha_\beta = 0.3$, $\sigma_\gamma = 1$, a convergence tolerance of 10^{-3} , and a maximum of 3,000 coordinate-ascent iterations. The COVID-19 single-cell simulations were fit at each of $d \in \{8, 10, 14\}$ latent programs, with the supervised-coefficient scale held fixed at $b_v = 1$. In the GTEx bulk simulations the number of simulated programs is $d^* = 8$, while the fitted models used $d = 10$ for the unmasked configuration, $d_{\text{path}} = 3$ for the pathway-masked configuration, and 3 pathway-anchored plus 7 *de novo* factors for the combined configuration; all unsupervised baselines were fit with $d = 10$ for comparability. In this arm b_v was initialized at 1 and auto-calibrated during fitting to the data precision, converging to ≈ 0.01 .

Table D1. Fixed parameters for the zero-mediation program-recovery and prediction benchmarks. Modality-specific settings and parameter sweeps are described in §3.2.

Symbol	Meaning	Single-cell	Bulk
–	individuals / samples	80 (100 cells each)	2,000
p	genes	10,000	5,000
–	cell types	6	–
–	disease programs (annot. + <i>de novo</i>)	2 + 2	2 + 4
–	nuisance programs (annot. + <i>de novo</i>)	0 + 1	1 + 1
$ S_\ell $	program support size	U{50,100}	U{50,150}
α_p, β_p	propensity Beta prior	2.0, 2.0	–
σ_u	per-program propensity jitter	0.5	–
(q_0, q_1)	baseline / max carrier prob.	(0.20, 0.80)	–
λ_ℓ	program prevalence multiplier	1.0 (all ℓ)	–
π_ℓ	nuisance carrier rate	0.5	0.5
ω_ℓ	liability weight per DRGP	{2.0, 0.7, 2.0, 0.7}, balanced signs	$ v = 1.5$
–	nuisance liability weight	0	0
–	liability form / prevalence	probit / 0.5	probit / 0.26
–	baseline activation magnitude	Gamma(2,2)	binary carrier
–	program polarity	up/up/mixed/down/mixed	up
–	truth draws per configuration	5	5

For experimental analyses, hyperparameters were selected separately for each dataset, input representation, cohort size, and prior-knowledge mode using tree-structured Parzen estimator optimization implemented in Optuna¹⁶. The objective was mean validation ROC-AUC. For single-cell analyses, all cells from the same patient were assigned to the same partition to prevent patient-level information leakage between training and validation sets. We evaluated 30 trials for each COVID-19 single-cell configuration and 60 trials for each COVID-19 pseudo-bulk and GTEx configuration.

The search included the number of latent programs d ; the Gamma prior shapes α_θ , α_β , α_ξ , and α_η ; the Laplace scale b_v for the supervised program coefficients; the prior standard deviation σ_γ for auxiliary-covariate coefficients; and the spike-and-slab parameters α_π and $\beta_{\pi,\text{scale}}$. For pathway-masked models, d was fixed to the number of supplied pathways. For combined models, the pathway-constrained factors were fixed and the total factor count was included in the hyperparameter search. The configuration with the highest validation AUC was selected. However, in the COVID-19 combined-mode cross-validation runs, a runtime override fixed the number of de novo factors at 50, yielding 346+50 factors for single-cell and 351+50 for pseudo-bulk analyses, irrespective of the selected total factor count. Final experimental models were run for at most 500 coordinate-ascent iterations with an ELBO-based relative convergence tolerance of 10^{-3} and a 50-iteration warm-up before updating the supervised coefficients. The selected values for each reported configuration are provided in Table D2. All comparison methods were run with default hyperparameters.

Table D2. **Hyperparameters used for the experimental configurations.** Values were selected using patient- or subject-grouped validation AUC on a single tuning split, subject to the COVID-19 combined-mode factor-count override described in Appendix §D; the same settings were held fixed across all 25 outer cross-validation folds. The number of latent gene programs is written $d_{\text{pathway}} + d_{\text{de novo}}$: unmasked models are entirely *de novo*, masked models are entirely pathway-anchored, and combined models use both. † Fixed at the default value 1.0 in combined mode.

Dataset/representation	Mode	$d_{\text{pathway}} + d_{\text{de novo}}$	α_θ	α_β	α_ξ	α_η	b_v	σ_γ	α_π	$\beta_{\pi,\text{scale}}$
COVID-19 single-cell	Unmasked	0 + 250	1.5193	2.6415	0.9882	1.5225	0.1932	0.3921	2.1577	2.1839
COVID-19 single-cell	Masked	346 + 0	2.3834	2.0881	3.8469	2.1605	1.6505	1.9670	0.8117	41.1046
COVID-19 single-cell	Combined	346 + 50	0.8097	1.1859	1.0000†	1.0000†	0.0591	0.2650	0.4575	2.8908
COVID-19 pseudo-bulk	Unmasked	0 + 50	2.4135	1.1470	3.8935	3.2130	1.5999	1.4595	1.0371	36.8330
COVID-19 pseudo-bulk	Masked	351 + 0	2.4135	1.1470	3.8935	3.2130	1.5999	1.4595	1.0371	36.8330
COVID-19 pseudo-bulk	Combined	351 + 50	0.6698	3.4088	1.0000†	1.0000†	0.0756	1.8726	0.6672	7.7920
GTEx whole blood	Unmasked	0 + 130	3.7266	0.8097	1.1859	0.6583	0.1660	0.3204	0.2891	25.5859
GTEx whole blood	Masked	309 + 0	1.4833	2.3994	0.9932	3.3077	0.0658	1.9230	2.0513	2.1758
GTEx whole blood	Combined	309 + 90	3.0180	2.7441	1.0000†	1.0000†	1.5048	1.0461	0.1240	27.2457

Appendix E. Additional Results

Appendix E.1. *Simulation Results Across the Parameter Sweep*

The single-cell simulation was run over the difficulty grid of §3.2 rather than a single configuration, summarised here as marginal profiles along each factor. Three patterns hold across the sweeps. Annotated recovery is set by whether a method is given the supports rather than by the difficulty of the instance: DRGP-masked is invariant at 0.801 and DRGP-combined spans 0.762–0.801, while methods without the supports stay near the chance level $|S_\ell|/p = 0.0075$ (Figs. E1, E2). *De novo* recovery is at chance for every method in every configuration (0.008–0.010). Patient-level prediction responds most to the responder fraction and to the transcriptomic liability share, and least to the perturbation strength: across the sweep the largest swing in AUC is 0.146 for ρ and 0.118 for w_Z , against at most 0.092 for r (Figs. E4, E3).

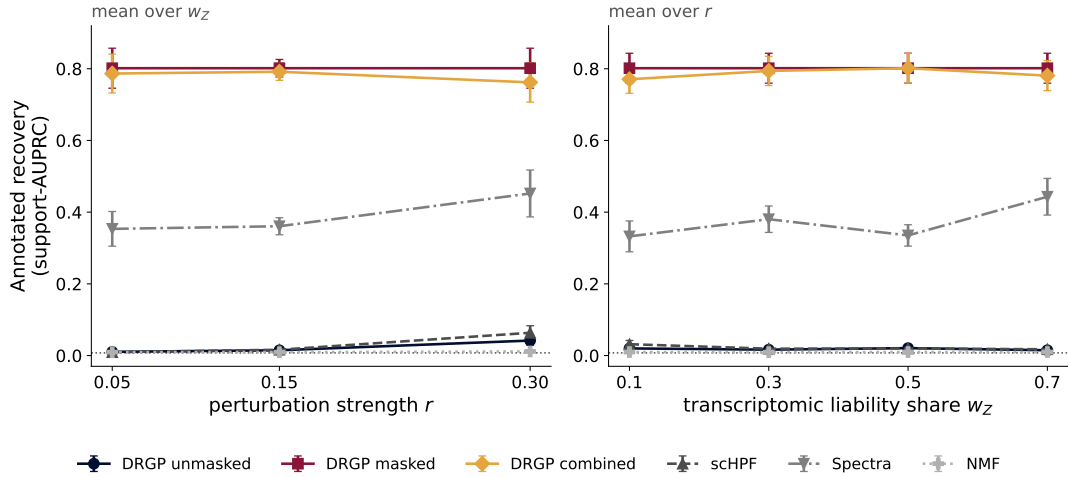


Fig. E1. **Recovery of Pathways (support-AUPRC) as a function of r and w_Z for a fixed $\rho = 0.3$.** Left: Support-AUPRC against r , averaging over $w_Z \in \{0.1, 0.3, 0.5, 0.7\}$. Right: Support-AUPRC against w_Z , averaging over $r \in \{0.05, 0.15, 0.30\}$. Both panels hold $\rho = 0.3$ and $d = 8$. Points are means over the five simulated truth draws with ± 1 standard error, and the dotted line marks chance ($|S_\ell|/p = 0.0075$).

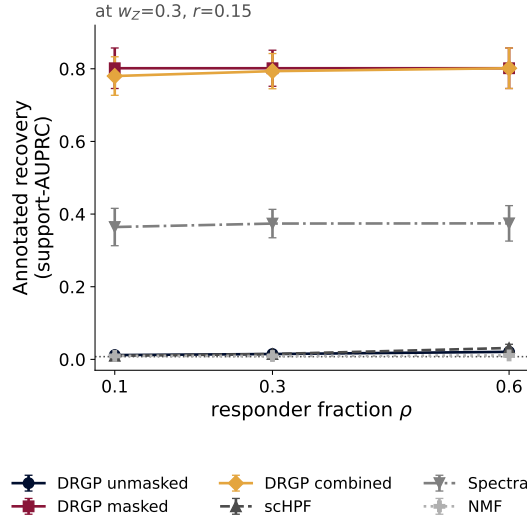


Fig. E2. **Recovery of Pathways (support-AUPRC) on the single-cell simulation as a function of ρ for a fixed $w_Z = 0.3$, $r = 0.15$, $d = 8$.** Points are means over five simulated-truth draws with ± 1 standard error; the dotted line marks chance, the mean support prevalence $|S_\ell|/p = 0.0075$.

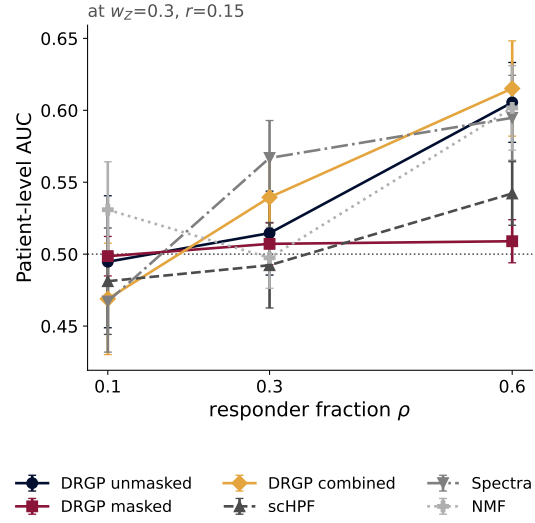


Fig. E3. **Patient-level AUC in the single-cell simulation as a function of ρ for a fixed $w_Z = 0.3$, and $r = 0.15$.** Points are means over five simulated-truth draws with ± 1 standard error; the dotted line marks chance.

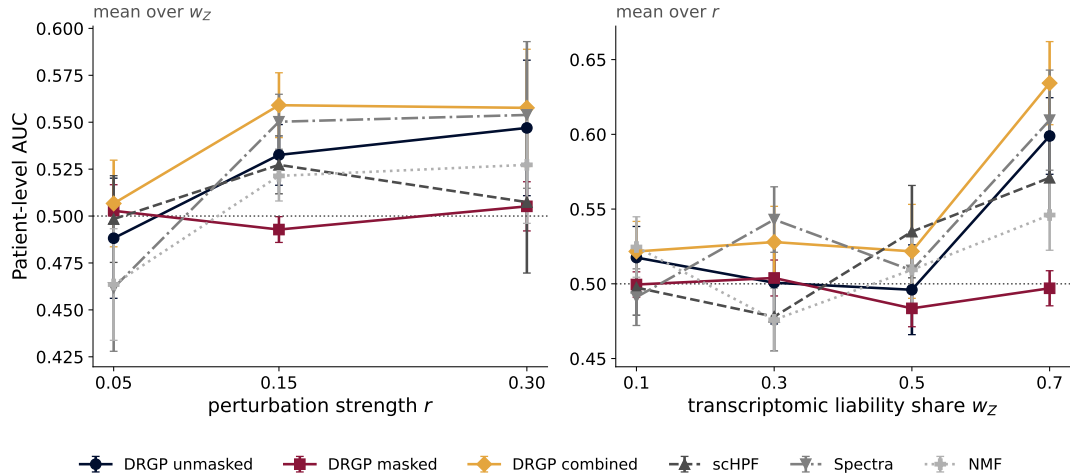


Fig. E4. **Patient-level AUC in the COVID-19 single-cell simulation as a function of r and w_Z for a fixed $\rho = 0.3$.** Left: AUC against r , averaging over $w_Z \in \{0.1, 0.3, 0.5, 0.7\}$. Right: AUC against w_Z , averaging over $r \in \{0.05, 0.15, 0.30\}$. Both panels hold the responder fraction at $\rho = 0.3$ and the latent dimension at $d = 8$. Points are means over the five simulated truth draws for each configuration, and error bars show ± 1 standard error. The dotted line marks chance AUC = 0.5.

Appendix E.2. *GTEx mediation calibration and evaluation*

We detail the coupling calibration and evaluation of the GTEx mediation simulations defined in §2.3. Here, $\tilde{s}_i$ is the standardized PRS, $a_{i\ell}^{(0)} = c_{i\ell}$, and individual-level program activities are $A_{i\ell} = a_{i\ell}$.

Coupling scale and interpretation. Write

$$g = \sum_{\ell \in \mathcal{D}} |\omega_\ell|, \quad \sigma_c^2 = \text{Var} \left(\sum_{\ell \in \mathcal{D}} \omega_\ell c_{i\ell} \right).$$

Under independence of the carrier indicators and standardized PRS, the variance of the PRS-dependent component entering liability through the program channel is

$$\frac{w_Z \kappa_{\text{med}}^2 g^2}{\sigma_c^2 + \kappa_{\text{med}}^2 g^2}.$$

Setting this quantity to $\tilde{m}w_s$ gives

$$\kappa_{\text{med}} = \frac{1}{g} \sqrt{\frac{\tilde{m}w_s \sigma_c^2}{w_Z - \tilde{m}w_s}}, \quad w_Z > \tilde{m}w_s. \quad (\text{E.1})$$

This calibration specifies the variances of the mediated and direct components separately. Because both components depend on the same PRS and act in the same liability direction, their combined variance is

$$\text{Var}(L_{\text{genetic}}) = w_s \left(\sqrt{\tilde{m}} + \sqrt{1 - \tilde{m}} \right)^2. \quad (\text{E.2})$$

Thus, w_s is a nominal genetic weight, and $\tilde{m}$ controls the allocation between pathways; intermediate values do not preserve total genetic variance. These experiments assess model behavior under PRS-expression dependence, rather than isolating mediation at a fixed total signal strength.

Simulation settings. The nine parameter combinations in §3.2 yield 45 simulations, each with 2,000 individuals, 5,000 genes, and target prevalence 0.26. Within each truth replicate, the three w_s conditions share the carrier design, PRS vector, and gene loadings. Background expression and PRS sampling follow §2.3.

Attribution and prediction diagnostics. For each fitted model, we measure the maximum absolute correlation between the PRS and any inferred program, $T = \max_\ell |\text{corr}(\hat{\theta}_\ell, \tilde{s})|$. A permutation reference is obtained from 50 permutations of the PRS across individuals, recomputing the statistic with the inferred programs held fixed. The count annotations in Fig. 5A indicate how many of the five truth replicates have an observed statistic exceeding their corresponding reference, separately for each $(\tilde{m}, w_s)$ combination. Conditions sharing a truth replicate are not independent. We also report the posterior mean auxiliary PRS coefficient $\hat{\gamma}$ and held-out prediction AUC, summarized by the mean and standard deviation across the five truth replicates per parameter combination. Program-PRS association measures whether inferred programs capture the introduced genetic dependence. The auxiliary coefficient describes the remaining predictive association conditional on inferred programs; it is not, by itself, an estimate of a causal direct effect.

Appendix E.3. *Repeated cross-validation on experimental data*

Five repeats of stratified five-fold cross-validation were generated at the individual level. All cells and biological samples from an individual remained together, and every method used the same saved partitions. Each individual appeared in the test partition once per repeat. Within each outer training partition, 15% of individuals were assigned to a stratified validation partition. Resulting training/validation/test sizes were 428–429/76/126–127 individuals for GTEx and 88–89/16/26–27 patients for both COVID-19 representations.

Single-cell metrics were calculated over individual cells using inherited patient labels; cell predictions were not averaged within patients for this evaluation. Consequently, patients contributing more cells contribute more observations to these metrics. Pseudo-bulk and GTEx metrics were calculated over individuals. ROC-AUC and average precision were computed separately for each fold and summarized across 25 folds.

All bulk representations were fitted on training partitions only. For scHPF and Spectra, gene programs were then frozen and held-out expression was projected onto them. In single-cell analyses, scHPF and Spectra representations were learned once from the full cohort expression matrix, with logistic classifiers fitted separately within each fold. Only these single-cell representation evaluations are transductive.

DRGP hyperparameters were selected once per cohort, representation, and prior-knowledge mode using patient-grouped validation splits, then held fixed across the repeated evaluation. Hyperparameter selection was not nested within outer folds, so performance estimates may be optimistic.

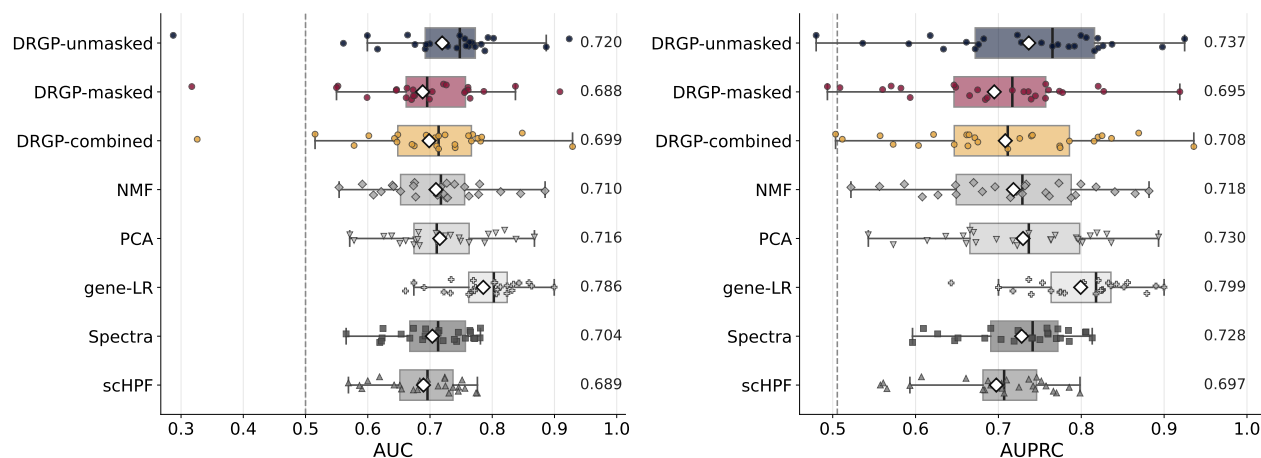


Fig. E5. **Repeated cross-validation performance on COVID-19 single-cell data.** ROC-AUC (left) and average precision (right) across 25 individual-grouped test folds. Boxes show medians and interquartile ranges, whiskers extend to the most extreme observations within $1.5\times$ the interquartile range, points denote fold scores, and white diamonds mark means. Fold scores are dependent because individuals recur across repeats and training partitions overlap. Single-cell metrics are evaluated per cell.

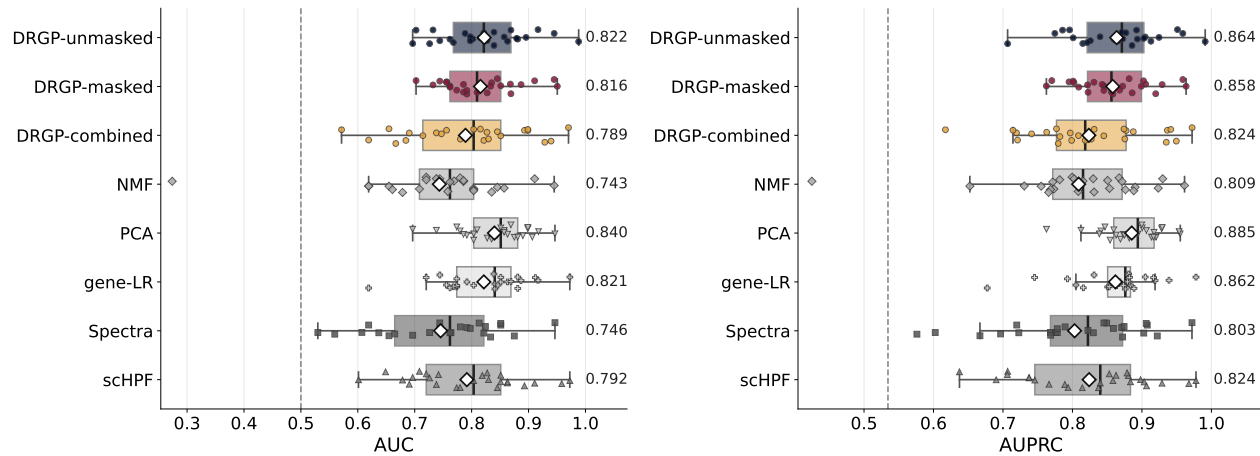


Fig. E6. **Repeated cross-validation performance on COVID-19 pseudo-bulk data.** ROC-AUC (left) and average precision (right), evaluated per patient across 25 test folds from five repeats of five-fold patient-grouped cross-validation. All methods use training-fold representation fits; scHPF and Spectra project held-out expression onto frozen gene programs. Boxes show medians and interquartile ranges, whiskers extend to observations within $1.5\times$ the interquartile range, and points show fold scores. White diamonds and numerical annotations indicate means. Fold scores are dependent because patients recur across repeats and training partitions overlap.

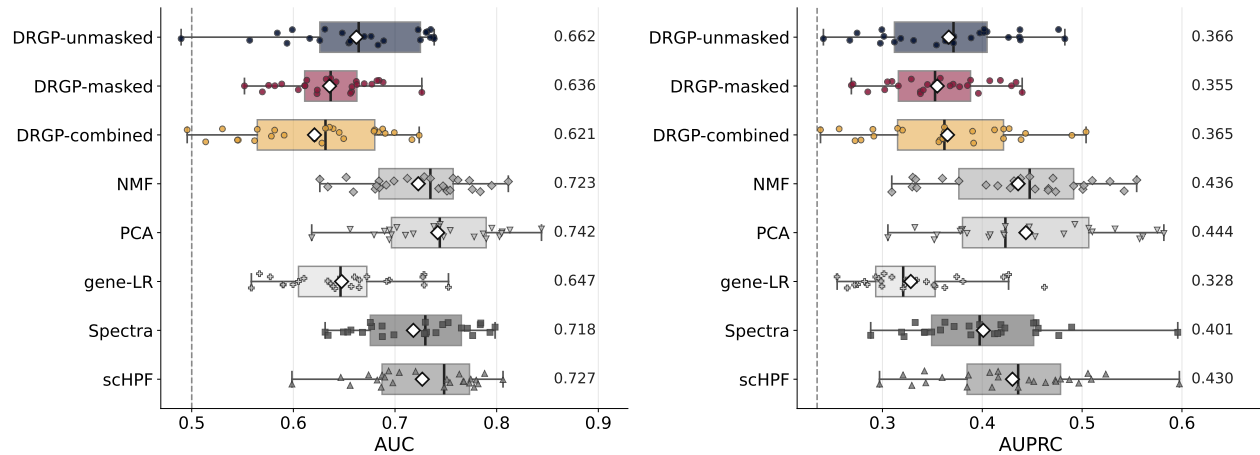


Fig. E7. **Repeated cross-validation performance on GTEx whole-blood data.** ROC-AUC (left) and average precision (right), evaluated per individual across 25 test folds from five repeats of five-fold individual-grouped cross-validation. Predictors include expression-derived features, clinical and demographic covariates, and the CAD PRS. All methods use training-fold representation fits; scHPF and Spectra project held-out expression onto frozen gene programs. Boxes show medians and interquartile ranges, whiskers extend to observations within $1.5\times$ the interquartile range, and points show fold scores. White diamonds and numerical annotations indicate means. Fold scores are dependent because individuals recur across repeats and training partitions overlap.

Appendix E.4. *Probability calibration*

We evaluated probability calibration on GTEx whole blood and COVID-19 representations using five repeats of stratified, individual-grouped five-fold cross-validation used for predictive evaluation. Analyses included all three DRGP configurations, NMF, PCA, gene-LR; bulk analyses additionally included Spectra, and scHPF.

For each fold, Platt scaling fitted logistic regression to the logits of validation probabilities and applied the fitted transformation to held-out test probabilities. Test observations were not used to fit the calibrator. Reliability diagrams compare mean predicted probability with observed positive frequency within ten equal-width probability bins, pooling held-out predictions across all five repeats. Bins containing fewer than ten prediction records were omitted. Each individual (bulk) or cell (single-cell) contributes one held-out prediction per repeat, so the pooled records are not independent. Single-cell summaries weight cells equally rather than patients equally; cells inherit their patient’s label, and patients contributing more cells receive greater weight.

COVID-19 results. For pseudo-bulk data, the Spectra and scHPF reliability curves were generally closer to the diagonal after Platt scaling, although deviations remained and changes varied across methods (Fig. E9). Single-cell reliability curves also retained substantial departures from the diagonal after calibration (Fig. E8).

GTEx results. Raw DRGP probabilities tended to overestimate observed disease frequencies. Platt scaling brought their reliability curves closer to the diagonal, but overestimation remained at higher predicted probabilities (Fig. E10).

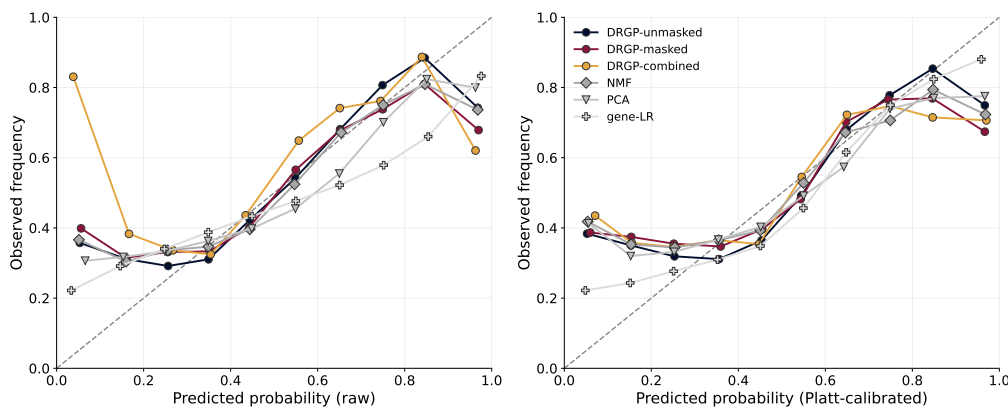


Fig. E8. **Probability calibration on COVID-19 single-cell data.** Raw probabilities (left) and probabilities after validation-fitted Platt scaling (right). Points summarize ten equal-width probability bins, pooling held-out cell predictions across five repeats of five-fold patient-grouped cross-validation. Bins with fewer than ten prediction records are omitted; the dashed diagonal denotes perfect calibration. The cohort contains 428,334 positive and 418,513 negative cells (50.6% positive). Cells inherit their patient’s label; patients are the independent sampling units, and patients contributing more cells receive greater weight in these curves. Each cell contributes one held-out prediction per repeat. Spectra and scHPF are omitted because paired validation and test predictions were unavailable.

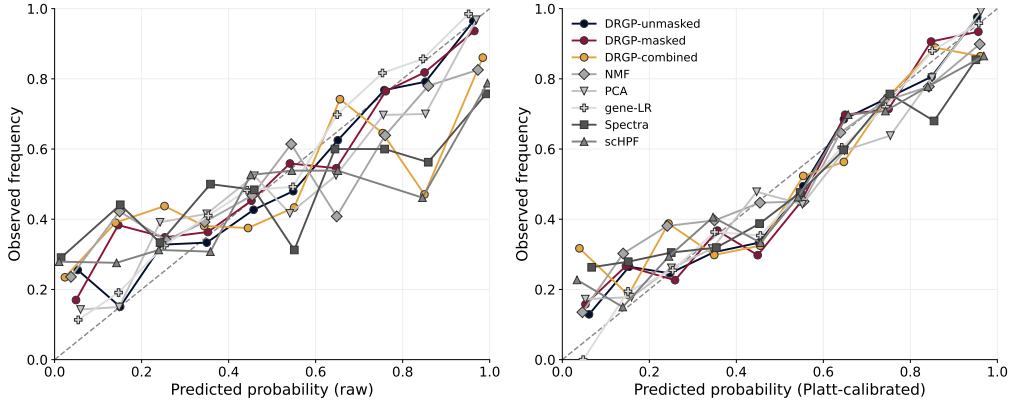


Fig. E9. **Probability calibration on COVID-19 pseudo-bulk data.** Raw probabilities (left) and probabilities after validation-fitted Platt scaling (right). Points show mean predicted probability and observed positive frequency within ten equal-width probability bins, pooling held-out predictions across five repeats of five-fold patient-grouped cross-validation. Bins with fewer than ten prediction records are omitted. The dashed diagonal denotes perfect calibration. The cohort contains 70 severe or critical and 61 mild or moderate patients; each patient contributes one held-out prediction per repeat.

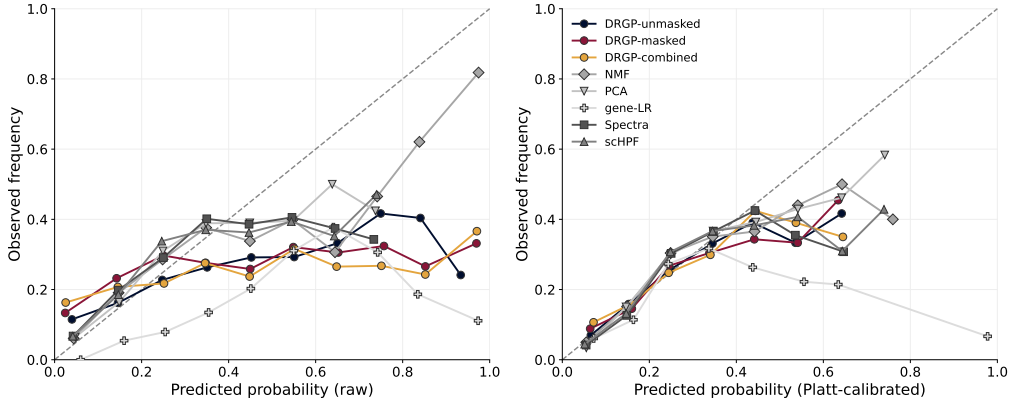


Fig. E10. **Probability calibration on GTEx whole blood data.** Raw probabilities (left) and probabilities after validation-fitted Platt scaling (right). Points show mean predicted probability and observed positive frequency within ten equal-width probability bins, pooling held-out predictions across five repeats of five-fold cross-validation. Bins with fewer than ten prediction records are omitted. The dashed diagonal denotes perfect calibration. The cohort contains 148 positive and 483 negative individuals; each individual contributes one held-out prediction per repeat.

Appendix E.5. *Cell-Level Prediction in the Single-Cell Simulation*

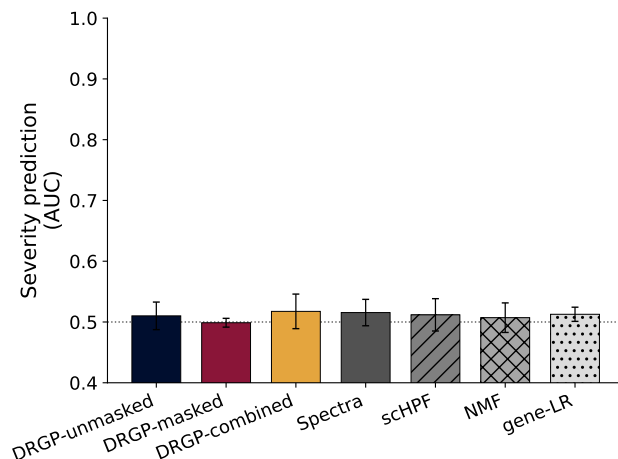


Fig. E11. **All methods remain at chance on simulated DPP.** COVID-19 scRNA-seq simulations evaluated on cells from held-out patients using inherited labels. Bars show means with error bars denoting ± 1 SD. Dotted lines mark chance performance.

Assigning each cell its individual’s phenotype produces a label that individual cells cannot support. At the headline setting of the simulated grid ($w_Z = 0.7$, $r = 0.30$), cell-level prediction remains at chance for every method (0.50–0.52 AUC, Fig. E11), not because the programs are undetectable, but because a responder fraction of $\rho = 0.3$ leaves most cells in a case individual carrying no DRGP at all. Averaging a patient’s cell-level probabilities recovers the signal that individual cells cannot resolve, making patient-level performance the more appropriate and reliable measure of predictive accuracy in this setting.

Appendix E.6. *Reproducible Gene Programs on Experimental Results*

The programs interpreted in §3.4 were selected by a cross-seed reproducibility filter, applied separately to each cohort.

Program interpretation used separate combined-mode fits with the configurations in Table E1. The GTEx interpretation fits included all 27,693 input genes without protein-coding restriction, whereas predictive evaluation used protein-coding genes only. These interpretation fits were used exclusively for program analysis and did not contribute to the reported predictive performance.

The two program classes require different matching rules. A pathway factor is basically identified by its name, which is constant across seeds, and only the ranking of genes within the mask can vary. A *de novo* program has no fixed identity and hence, those programs are grouped into cross-seed clusters, where each factor joins an existing cluster with the highest mean Jaccard to its members, provided that the mean is at least 0.30, and otherwise seeds a new cluster. A program cluster is then called reproducible when it is present in at least 3 of the five initialization seeds (Fig.E12).

Table E1. **Configurations used for program interpretation.** Both cohorts were fit in *combined* mode with five random seeds, varying only the model-initialization seed. GTEx interpretation experiments included protein-coding and noncoding genes; predictive analyses used protein-coding genes only. d is written $d_{\text{path}} + d_{\text{de novo}}$. Values not listed were left at their defaults; in particular α_ξ and α_η were not tuned for these runs.

	GTEx whole blood	COVID-19 pseudo-bulk
Label	heart disease	severity
Input	bulk counts, 27,693 genes	pseudo-bulk counts, 131 patients
Pathway mask	Reactome curated (depth 2, ≥ 15)	Reactome immune
$d_{\text{pathway}} + d_{\text{de novo}}$	309 + 300	358 + 50
Auxiliary covariates	5 (sex, BMI, smoking, hypertension, T2D)	sex, 3 comorbidities
α_θ	0.6814	2.4135
α_β	2.0062	1.1470
α_ξ	1.0 (default)	1.0 (default)
α_η	1.0 (default)	1.0 (default)
b_v	0.2413	1.5999
σ_γ	1.2273	1.4595
α_π	0.2407	1.0371
$\beta_{\pi, \text{scale}}$	5.1838	36.8330
Max. CAVI iterations	500	500
Relative tolerance	10^{-3}	10^{-3}
Supervised warm-up	50 iterations	50 iterations
Early stopping	ELBO	ELBO
Regression design	normalized	standard

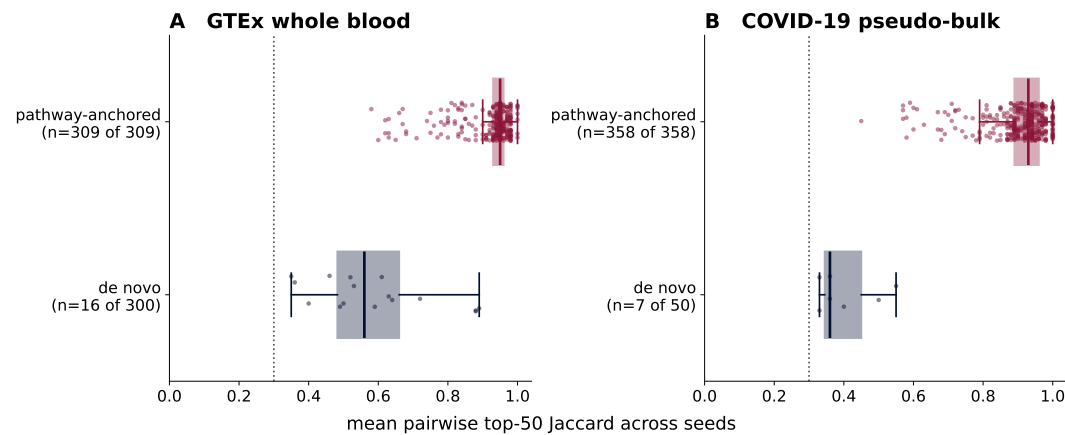


Fig. E12. **Cross-seed stability of the interpreted gene programs.** Mean pairwise Jaccard between the top-50 gene sets of a program’s members across five random seeds, for **(A)** GTEx whole blood (309 pathway-anchored + 300 free factors), and **(B)** COVID-19 pseudo-bulk (358 pathway-anchored + 50 free factors), both in combined mode. Boxes show the median and interquartile range with whiskers to $1.5\times\text{IQR}$; points are individual programs. The dotted line marks the selection threshold ($J \geq 0.30$, together with presence in ≥ 3 of 5 seeds). Axis labels give the number of programs retained out of the number available in each class.

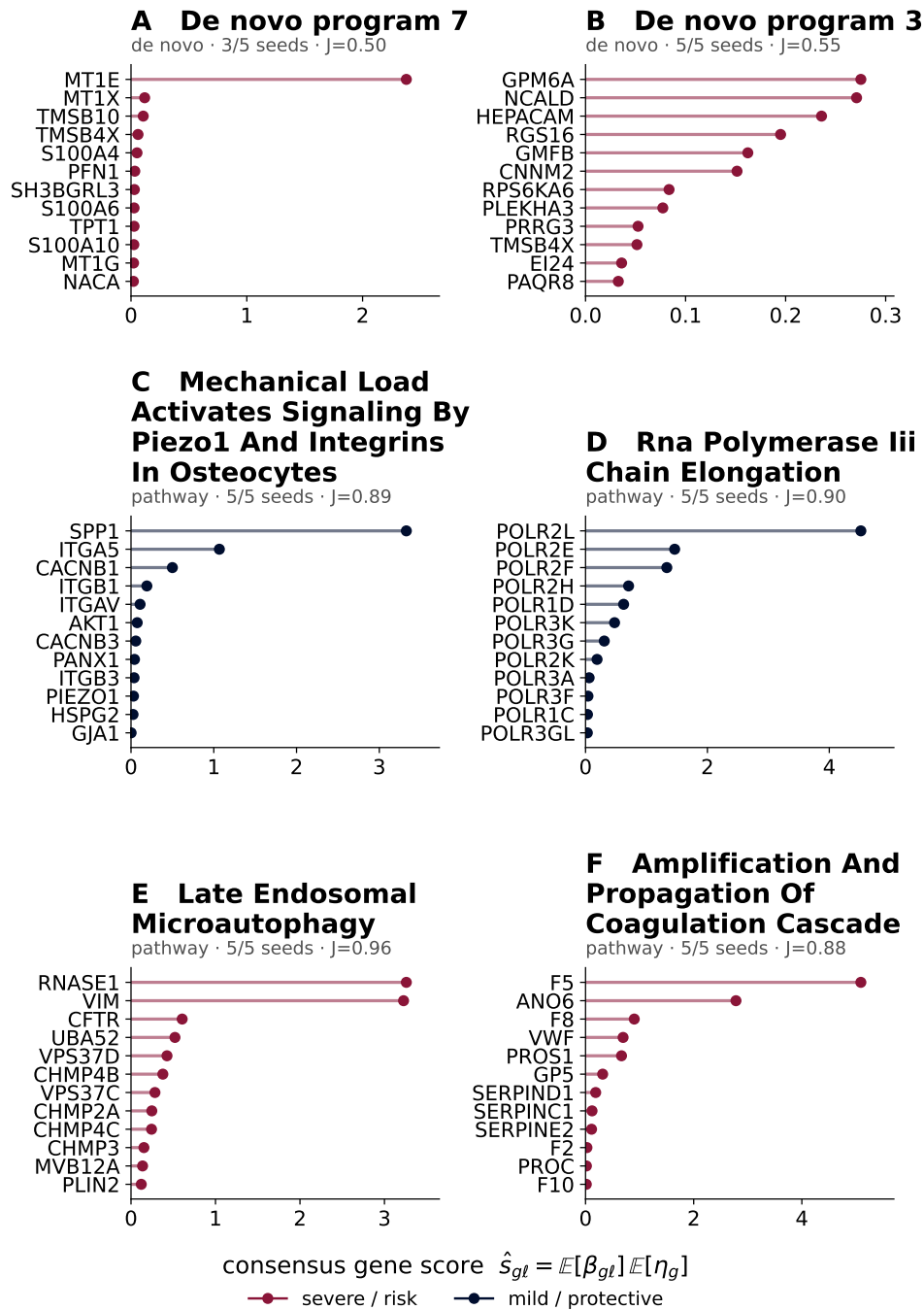


Fig. E13. **Reproducible COVID-19 gene programs and their leading genes.** The six programs with the largest effect-scaled importance $I_\ell = |\bar{v}_{k\ell}| \text{sd}(\bar{\theta}_\ell)$ among those passing the reproducibility filter (Appendix §E.6), from the pseudo-bulk combined fit with the Reactome curated mask. Each panel shows the top 12 genes of the consensus signature, ranked by the mean score $\hat{s}_{gl}$ across the seeds in which the program recurs. Panel headers give the program kind (*de novo* GP or pathway), the number of seeds in which it was found, and its mean pairwise top-50 Jaccard. Colour gives the sign of the supervised coefficient: maroon for programs weighted towards severe disease, navy for programs weighted towards mild disease.

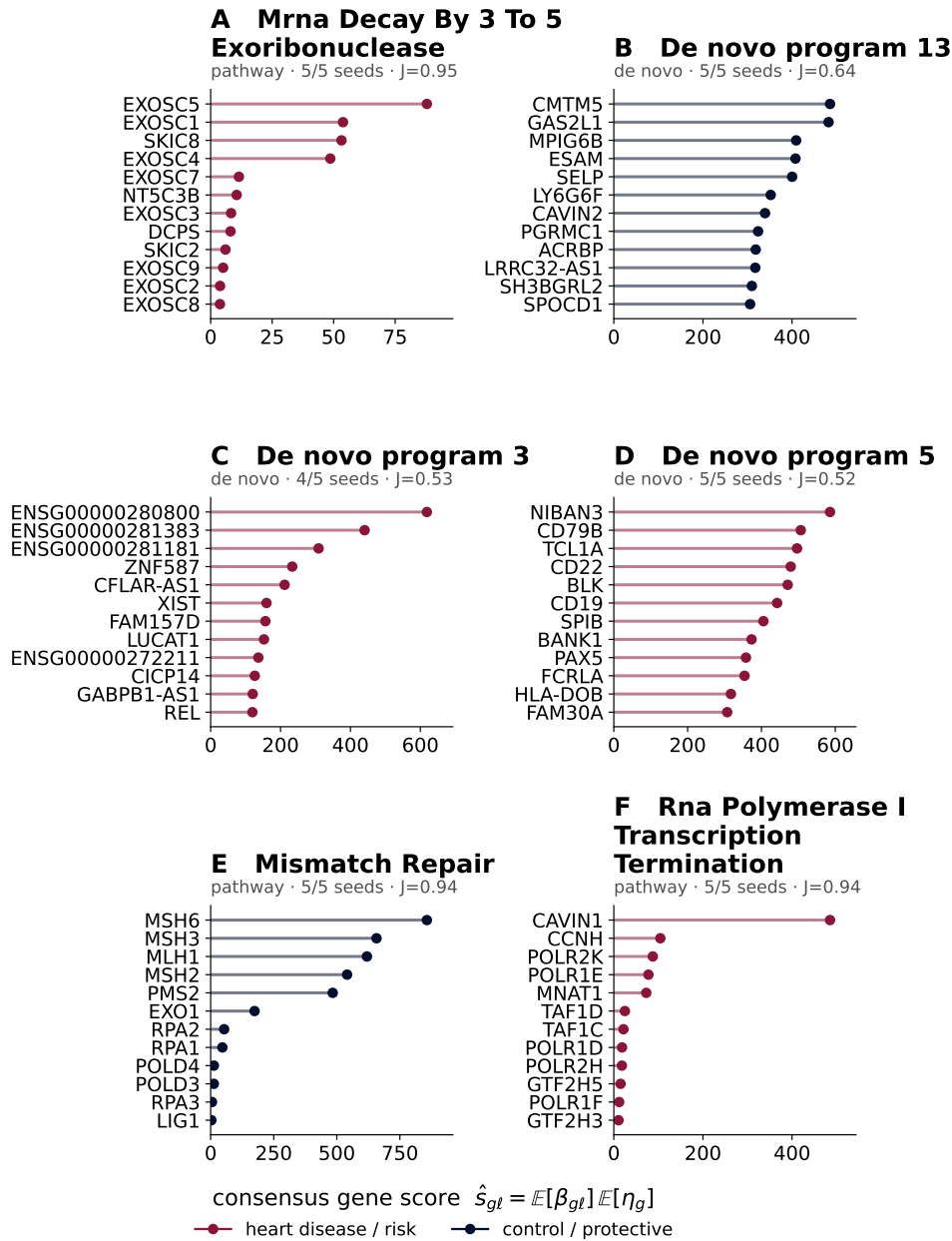


Fig. E14. **Reproducible GTEx heart disease gene programs and their leading genes.** The six programs with the largest effect-scaled importance I_ℓ among those passing the reproducibility filter(Appendix §E.6), from the GTEx heart disease combined fit with the Reactome curated mask. Panel headers give the program kind (*de novo* GP or pathway), the number of seeds in which it was found, and its mean pairwise top-50 Jaccard. Colour gives the sign of the supervised coefficient: maroon for programs weighted towards heart disease cases, navy for programs weighted towards controls.

For each reproducible program we report a consensus top-50 list, ranking genes by their mean $\hat{s}_{gl}$ across seeds in which the program appears. Ensembl identifiers that could not be mapped to a gene symbol are reported as identifiers.

Table E2. Disease-related genes within disease-associated gene programs.

Program	Dataset	Interpretation	Genes
GP7	COVID-19	Cell proliferation	<i>MKI67, TOP2A, CKS2, ZWINT, CENPF, CEP55, CCNB2, CDKN3, NCAPH, NUSAP1, PKMYT1</i>
GP7	COVID-19	Monocyte and macrophage markers	<i>MT1X, MT1G, S100A4, S100A6, LGALS1, COTL1, SRGN, HMGN2</i>
GP7, GP3	COVID-19	Immune-regulatory metallothionein proteins	<i>MT1A, MT1E, MT1F, MT1G, MT1H, MT1M, MT1X</i>
GP7, GP3	COVID-19	Proteins that require binding to zinc or other metals to function	<i>ADNP, ARIH1, CDCA7, MAST4, NCALD, PAQR8, PKMYT1, PPM1N, PSPH, RPS6KA6, S100A4, S100A6, TOP2A, TRIM13, ZBTB20</i>
GP13	GTEEx	Cell adhesion	<i>CLDN5, EGFL7, MMRN1, SELP, TGFB1I1</i>
GP3	GTEEx	Megakaryocyte or platelet markers	<i>CAVIN2, CMTM5, GP6, PEAR1, SELP</i>
GP5	GTEEx	B cell markers	<i>BANK1, BLK, CD19, CD22, CD72, CD79B, FCRL1, FCRL2, FCRL5, FCRLA, HLA-DOB, IGHD, NIBAN3, PAX5, PLPPR5, TCL1A, VPREB3</i>

All pathway-masked programs pass the filter in both cohorts. This is expected rather than informative: because masked entries are held at $m_{gl} = 0$ and excluded from all updates (§B), a masked program’s gene support cannot change between seeds, and the Jaccard statistic measures only the stability of the within-mask ranking. The reproducibility result that carries information is therefore the *de novo* one, and it is far more selective.

Appendix F. Runtime Analysis

Hardware and software. The experimental prediction analyses ran on a heterogeneous SLURM cluster. DRGP and single-cell Spectra used one GPU per job (NVIDIA A100 80 GB, L40S 48 GB, or A10 24 GB). Bulk Spectra, scHPF, NMF, PCA, and gene-LR ran on CPUs. DRGP and NMF/PCA/gene-LR jobs were allocated eight CPU cores; scHPF and Spectra jobs were allocated four. The software environment included Python 3.10.18, JAX/jaxlib 0.6.2, NumPy 1.26.4, SciPy 1.15.2, scikit-learn 1.7.1, pandas 2.3.1, anndata 0.11.4, and scHPF v0.3.0-alpha (commit [aff30d6](#)). DRGP computations used float32 precision.

Runtime and memory. Figure F1 and Table F1 summarize job wall-clock times and peak host memory from scheduler accounting. For GTEEx and COVID-19 pseudo-bulk, all methods were fitted separately on each of the 25 training partitions. For scHPF and Spectra, the fitted gene programs were frozen before projecting held-out expression. For COVID-19 single-cell data, scHPF and Spectra representations were fitted once on the full cohort, with fold-specific

logistic classifiers; other methods were refitted per fold. Cohort sizes shown in the figure are total sizes, whereas per-fold fits use training subsets.

For each method and fold, accounting retained the longest recorded run, including when a fold was rerun. For bulk scHPF and Spectra, retained runs may therefore include additional classifier variants. These measurements describe individual job workloads, not cumulative computation across all attempts. Hardware and training-set sizes differ across jobs, so the timings are not a controlled comparison of algorithmic efficiency.

Median GTEx job times were 5–8 min for DRGP, 2 min for Spectra, and 14 min for scHPF; corresponding COVID-19 pseudo-bulk times were 2–3 min, 1.5 min, and 3.5 min. All reported bulk jobs used less than 5 GB of peak host memory. Single-cell DRGP median job times were 251–568 min, with recorded peak host memory of 59–112 GB. The full-cohort single-cell Spectra and scHPF jobs took 649 and 2,006 min and reached 58 and 561 GB, respectively; scHPF was allocated 900 GB. GPU-memory usage was not measured.

Random seeds. For repeat r and fold f , with $r, f \in \{0, \dots, 4\}$, outer partitions used seed r and within-training validation partitions used seed $100r + f$. All methods read the same stored partition files. DRGP and the standard baselines used initialization seed 0, and Spectra used seed 42. Additional initialization runs for repeat 0 used seeds 2001–2004 and were excluded from the headline cross-validation means. PyTorch was explicitly seeded for the bulk Spectra runs.

Pathway-mask construction. For predictive GTEx analyses, we used human Reactome release 96¹⁷, excluding the *Disease* and *Drug ADME* branches and their descendants. The retained collection comprised 309 depth-2 pathways with at least 15 member genes, spanning 5,745 distinct genes before intersection with the expression data. Pathway sizes ranged from 15 to 104 genes (median 41), and pathways retained their Reactome stable identifiers.

For predictive COVID-19 analyses, we used a fixed selection of 500 pathways from human MSigDB v2024.1, collection C2:CP¹⁸. The selection contained 222 WikiPathways¹⁹, 103 Reactome²⁰, 96 KEGG²¹, 54 PID²², and 20 BioCarta²³ gene sets, plus five legacy signaling sets, spanning 13,398 distinct genes before intersection with the expression data.

For both collections, member gene symbols were mapped to Ensembl identifiers using the same mapping procedure as the expression matrices. Membership was restricted to retained expression genes, and pathways with no remaining members were removed. The resulting masks contained 309 pathways for GTEx, 351 for COVID-19 pseudo-bulk, and 346 for COVID-19 single-cell analyses. In masked mode, the number of programs equalled the number of retained pathways. The exact pathway collections are distributed as GMT files with the code.

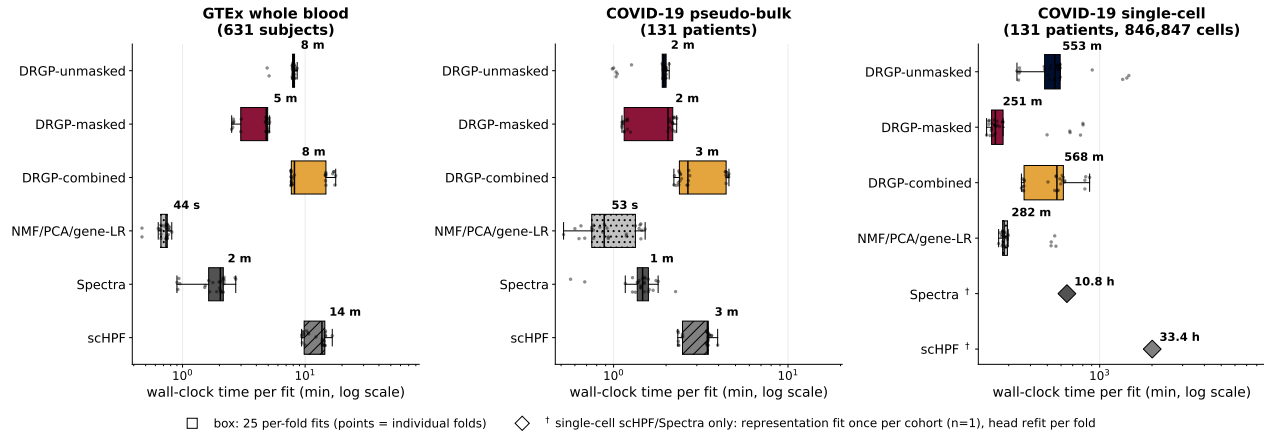


Fig. F1. Wall-clock times for experimental analyses. Times are shown on a logarithmic scale. Boxes summarize 25 per-fold jobs, with points showing individual folds. Diamonds (†) denote single full-cohort representation fits for Spectra and scHPF in the COVID-19 single-cell panel only; their logistic classifiers were refitted per fold. Both bulk panels use training-fold representation fits. Annotations show median times for repeated jobs and observed times for single jobs. Panel titles give total cohort sizes, not training-fold sizes. For repeated executions of a method–fold combination, the longest recorded run was retained. Hardware and workload differences preclude a controlled comparison of computational efficiency.

Table F1. **Compute environment and cost.** Median wall-clock time per job and scheduler-recorded peak host memory for the experimental prediction analyses. n denotes the number of contributing jobs. On both bulk cohorts every method, including Spectra and scHPF, is refit inside each of the 25 training folds. On the single-cell cohort only, the Spectra and scHPF representations were fitted once per cohort and just their fold-specific classifiers were refit; those classifier jobs are not included in the representation-job counts. Jobs were scheduled across heterogeneous GPUs (NVIDIA A100 80 GB, L40S 48 GB and A10 24 GB), so GPU times pool more than one device model. GPU memory was not measured.

Cohort	Method	n	Device	CPUs	Time/job	Host RAM
GTEx whole blood (631 individuals)	DRGP-unmasked	25	GPU	8	8 min	2.9 GB
	DRGP-masked	25	GPU	8	5 min	3.7 GB
	DRGP-combined	25	GPU	8	8 min	4.1 GB
	NMF/PCA/gene-LR	25	CPU	8	44 s	0.5 GB
	Spectra	25	CPU	4	2 min	1.4 GB
	scHPF	25	CPU	4	14 min	4.0 GB
COVID-19 pseudo-bulk (131 patients)	DRGP-unmasked	25	GPU	8	2 min	2.5 GB
	DRGP-masked	25	GPU	8	2 min	2.5 GB
	DRGP-combined	25	GPU	8	3 min	3.6 GB
	NMF/PCA/gene-LR	25	CPU	8	53 s	0.7 GB
	Spectra	25	CPU	4	1.5 min	1.2 GB
	scHPF	25	CPU	4	3.5 min	1.3 GB
COVID-19 single-cell (131 patients, 846,847 cells)	DRGP-unmasked	25	GPU	8	553 min	104 GB
	DRGP-masked	25	GPU	8	251 min	112 GB
	DRGP-combined	25	GPU	8	568 min	59 GB
	NMF/PCA/gene-LR	25	CPU	8	282 min	249 GB
	Spectra	1	GPU	4	649 min	58 GB
	scHPF	1	CPU	4	2,006 min	561 GB

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